

Project report on

Global Cyberattack Trends: Data

Visualization and Processing



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Objectives of the Project

The primary objective of this project is to analyze global cybersecurity threats using Python-based data handling and visualization techniques.

The project aims to provide a comprehensive understanding of how cyberattacks affect different countries, industries, and users by examining real or simulated datasets.

Through this study, students learn how to read, clean, and process CSV files, display structured tabular data, and create meaningful graphs that reveal patterns, trends, and distributions of cyber incidents. Another important objective is to develop the skills required for interpreting cybersecurity data through bar graphs, line graphs, and pie charts, enabling a clearer understanding of which areas are most vulnerable and how financial losses have evolved over time.

The project also focuses on helping students understand the significance of data-driven decision-making in the modern digital world and emphasizes the importance of cybersecurity awareness.

This project strengthens analytical thinking, Python programming skills, and the ability to present technical information in a structured, visual, and easy-to-understand format.

Overall,

1. To analyze global cybersecurity incidents using Python.
2. To visualize data patterns using graphs.
3. To study trends, predictions, and insights from data.
4. To develop a secure login-based menu program.
5. To demonstrate filtering, sorting, and reporting features.

Introduction

Cybersecurity has become one of the most critical areas of concern in today's digital world, where almost every activity from banking and business to education and entertainment, depends on the internet.

With the rapid increase in online services, the number of cyberattacks has also grown, affecting millions of users and leading to significant financial, personal, and organizational losses.

This project focuses on understanding these cybersecurity threats through data analysis and visualization using Python.

By working with datasets that represent attack types, affected countries, financial losses, and industry-based incidents, the project aims to provide a clear picture of how cyber threats spread and which areas are most vulnerable.

Python libraries such as pandas and matplotlib are used to process the data and generate graphs that help simplify complex information.

The introduction of visual tools like bar graphs, line graphs, and pie charts allows for easy comparison and interpretation of trends over time.

This project not only builds technical skills in coding and data handling but also enhances awareness about the importance of cybersecurity in everyday life. Through this practical analysis, students gain valuable insights into the digital threats around them and learn the significance of adopting safe online practices.

Hardware and software requirements

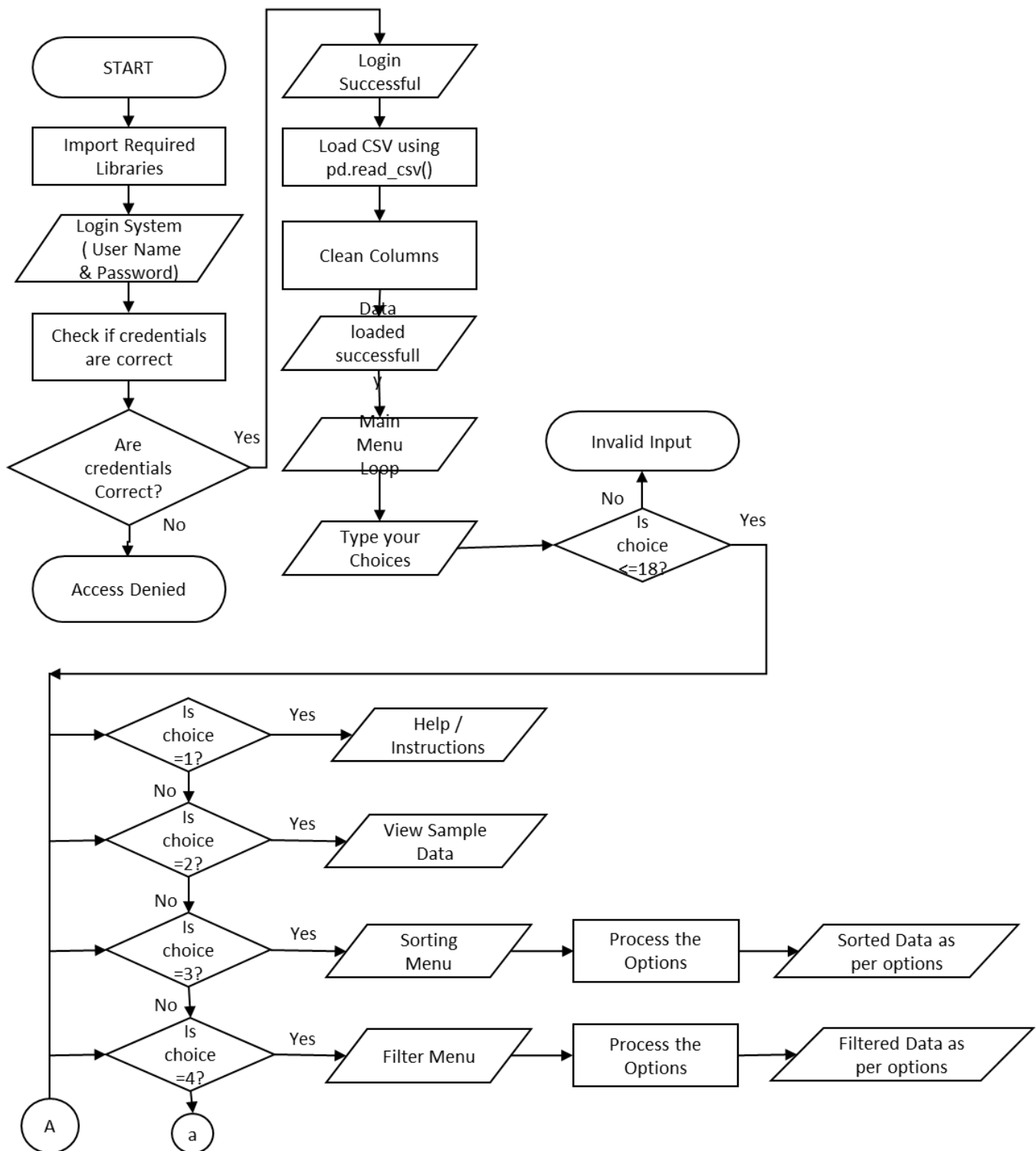
Hardware:

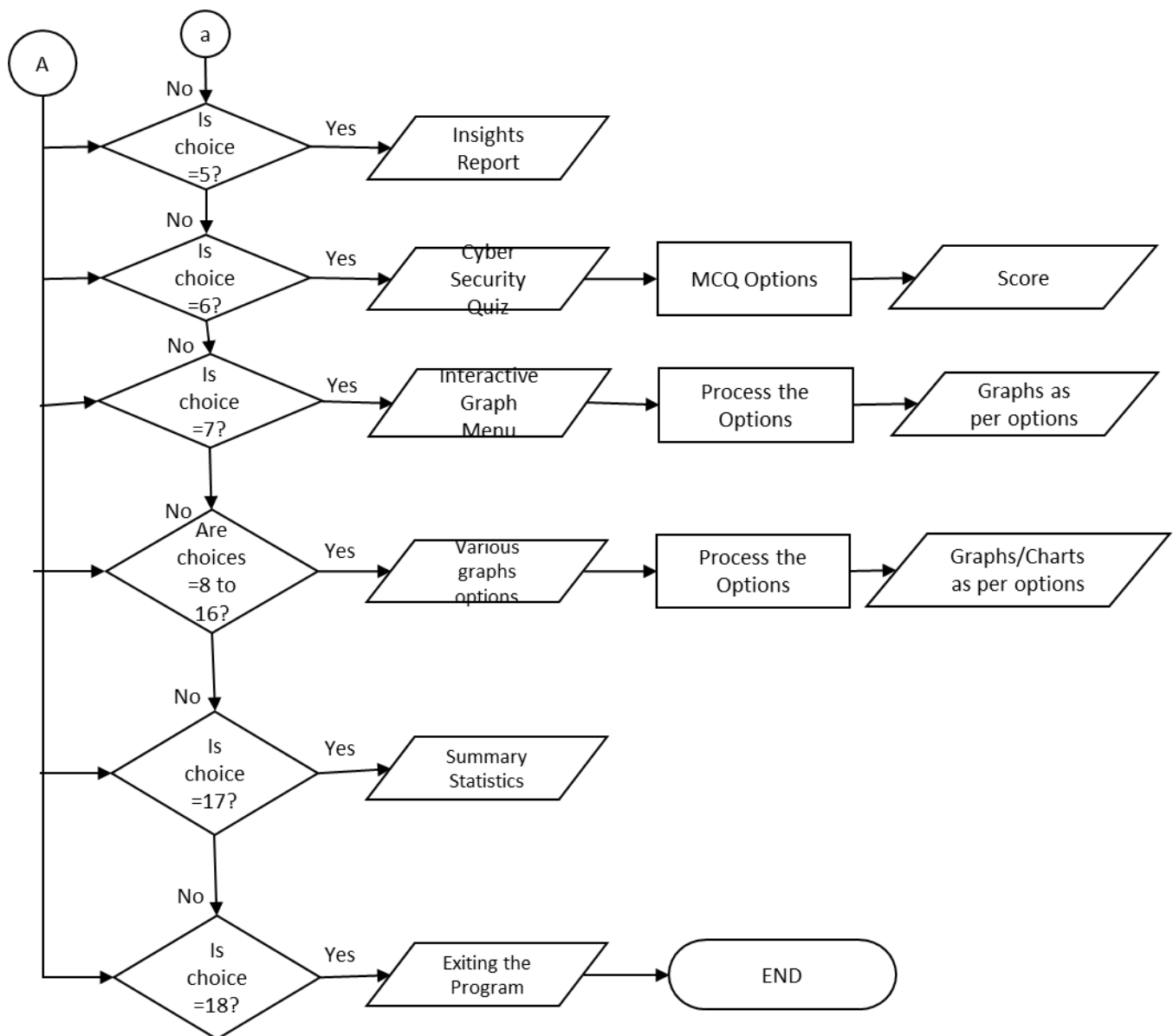
- Processor: 11th Gen Intel(R) Core(TM) i3-1115G4 @ 3.00GHz 2.90 GHz
- RAM: 8.0 GB
- System Architecture: 64-bit operating system, x64-based processor

Software:

- Operating system: Windows 10
- Spyder (Anaconda 3) version 3.8

Flow Chart





Libraries Used

The below are libraries used to complete this project.

- Pandas
- Matplotlib
- CSV

Table and Fields for CSV Files

Field Name	Datatype	Description
Year	Integer	Integer The year of the cyber incident (2015–2024).
Country	String	Country where the incident occurred.
Attack Type	String	Type of attack (Phishing, Ransomware, Malware, DDoS etc.).
Target Industry	String	Industry affected (Banking, Healthcare, Govt., IT etc.)
Financial Loss (in Million \$)	Float	Estimated loss in million dollars.
Number of Affected Users	Integer	Total affected users.

Derived Field	Datatype	Purpose
Country_clean	String	For lowercase, cleaned filtering.
Attack_clean	String	For cleaned attack type matching.
Industry_clean	String	For cleaned industry input.
Year_clean	Integer	Ensures year is numeric for filtering.

Source Code

```
import pandas as pd
import matplotlib.pyplot as plt

# ----- LOGIN SYSTEM -----
print("===== LOGIN REQUIRED =====")
username = "admin"
password = "123"

u = input("Enter Username: ")
p = input("Enter Password: ")

if u == username and p == password:
    print("\n✓ Login Successful!\n")
else:
    print("✗ Access Denied!")
    exit()

# ----- LOAD DATA -----
df =
pd.read_csv("C:\\Users\\Hp\\Downloads\\Global_Cybersecurity_Threats_2015-2024
(1).csv")
print("✓ Data Loaded Successfully!\n")

# Strip spaces from column names
df.columns = df.columns.str.strip()

# Clean columns for safe filtering
df["Country_clean"] = df["Country"].astype(str).str.strip().str.lower()
df["Attack_clean"] = df["Attack Type"].astype(str).str.strip().str.lower()
df["Industry_clean"] = df["Target Industry"].astype(str).str.strip().str.lower()
df["Year_clean"] = pd.to_numeric(df["Year"], errors='coerce')

# ----- MAIN MENU LOOP -----
```

```

while True:
    print("""
===== MAIN MENU =====
1. Help / Instructions
2. View Sample Data
3. Sorting Options
4. Filter Data
5. Insights / Auto Report
6. Cybersecurity Quiz
7. Interactive Graphs
8. Bar Graph - Country vs Number of Affected Users
9. Line Graph - Year vs Average Financial Loss
10. Pie Chart - Attack Type Distribution
11. Bar Graph - Industry vs Number of Incidents
12. Trend Analysis - Attack Types (Year-wise)
13. Trend Analysis - Industries (Year-wise)
14. Trend Analysis - Financial Loss (Year-wise)
15. Trend Prediction - Next Year Financial Loss
16. Overall Trend Analysis
17. Summary Statistics
18. Exit
""")
    choice = input("Type your choice: ").strip()

    # ----- 1. HELP / INSTRUCTIONS -----
    if choice == "1":
        print("""
===== HELP / INSTRUCTIONS =====

```

This project analyzes global cybersecurity data using:

- Menu-driven Python program
- Sorting, Searching, Filtering
- Data Visualization (Graphs)
- Statistical Summary

- Cybersecurity Quiz
- Auto-generated Reports

You can:

- View sample data
- Sort data (high → low or low → high)
- Filter by multiple conditions
- View interactive graphs
- Test your cybersecurity knowledge
- Generate insights/report

```
=====
```

```
""")
```

```
# ----- 2. SAMPLE DATA -----
```

```
elif choice == "2":
```

```
    print("\n===== SAMPLE DATA =====\n")
```

```
    print(df.head(10))
```

```
# ----- 3. SORTING -----
```

```
elif choice == "3":
```

```
    print("""
```

```
===== SORTING MENU =====
```

```
1. Sort by Financial Loss (High → Low)
```

```
2. Sort by Affected Users (High → Low)
```

```
3. Sort by Year (Old → New)
```

```
""")
```

```
    s = input("Type your choice: ").strip()
```

```
    if s == "1":
```

```
        print(df.sort_values("Financial Loss (in Million $)", ascending=False).head(10))
```

```
    elif s == "2":
```

```
        print(df.sort_values("Number of Affected Users", ascending=False).head(10))
```

```
    elif s == "3":
```

```
        print(df.sort_values("Year", ascending=True).head(10))
```

```
    else:
```

```

print("❌ Invalid Choice")

# ----- 4. FILTER DATA (UPDATED) -----
elif choice == "4":
    while True:
        print("\n===== FILTER MENU =====")
        print("1. Industry & Attack Type Filter")
        print("2. Country-wise Yearly Summary")
        print("3. Back to Main Menu")

        filter_choice = input("\nType your choice: ").strip()

        # ----- OPTION 1: Industry & Attack Type Filter -----
        if filter_choice == "1":
            print("\nAvailable Attack Types:")
            print(['DDoS', 'Malware', 'Man-in-the-Middle', 'Phishing', 'Ransomware', 'SQL
Injection'])
            print("\nAvailable Industries:")
            print(['Banking', 'Education', 'Government', 'Healthcare', 'IT', 'Retail',
'Telecommunications'])

            attack_input = input("\nType Attack Type (or press Enter to skip):
").strip().lower()
            industry_input = input("Type Industry (or press Enter to skip): ").strip().lower()

            filtered_df = df.copy()
            if attack_input:
                filtered_df = filtered_df[df["Attack_clean"] == attack_input]
            if industry_input:
                filtered_df = filtered_df[df["Industry_clean"] == industry_input]

            if filtered_df.empty:
                print("\n❌ No matching records found.")
            else:

```

```

print("\n✓ FILTERED DATA:\n")
print(filtered_df[[
    "Country", "Year", "Attack Type", "Target Industry",
    "Financial Loss (in Million $)", "Number of Affected Users"
]])

# ----- OPTION 2: Country-wise Yearly Summary -----
elif filter_choice == "2":
    print("\nAvailable Countries:")
    print(sorted(df["Country"].dropna().unique()))
    print("\nAvailable Years:")
    print(sorted(df["Year"].dropna().unique()))

    country_input = input("\nType Country (or press Enter to skip):
").strip().lower()
    year_input = input("Type Year (or press Enter to skip): ").strip()

    filtered_df = df.copy()
    if country_input:
        filtered_df = filtered_df[df["Country_clean"] == country_input]
    if year_input.isdigit():
        filtered_df = filtered_df[df["Year_clean"] == int(year_input)]

    if filtered_df.empty:
        print("\n✗ No matching records found.")
    else:
        print("\n✓ FILTERED DATA:\n")
        print(filtered_df[[
            "Country", "Year", "Attack Type", "Target Industry",
            "Financial Loss (in Million $)", "Number of Affected Users"
        ]])

# ----- OPTION 3: BACK TO MAIN MENU -----
elif filter_choice == "3":
    print("Returning to Main Menu...")

```

```

        break

    else:
        print("❌ Invalid choice! Please try again.")

# ----- 5. AUTOMATIC INSIGHTS -----
elif choice == "5":
    print("\n===== CYBERSECURITY INSIGHTS REPORT =====")

    top_country = df.groupby("Country")["Number of Affected Users"].sum().idxmax()
    top_attack = df["Attack Type"].value_counts().idxmax()
    top_year = df["Year"].value_counts().idxmax()
    max_loss_year = df.groupby("Year")["Financial Loss (in Million $)"].sum().idxmax()

    print("Most Attacked Country:", top_country)
    print("Most Common Attack Type:", top_attack)
    print("Year with Most Attacks:", top_year)
    print("Year with Highest Financial Loss:", max_loss_year)

print("\n=====")

# ----- 6. CYBERSECURITY QUIZ -----
elif choice == "6":
    print("\n===== CYBERSECURITY QUIZ =====")
    score = 0

    print("\n1) What is Phishing?")
    print("a) Fake emails to steal data")

    print("b) Hardware failure")
    print("c) Strong password")
    ans = input("Type your answer: ").lower()
    if ans == "a":

```

```
score += 1
```

```
print("\n2) What does Ransomware do?")
print("a) Displays ads")
print("b) Locks your data for ransom")
print("c) Cleans virus")
ans = input("Type your answer: ").lower()
if ans == "b":
    score += 1
```

```
print("\n3) Strong password includes?")
print("a) Only letters")
print("b) Letters, numbers, symbols")
print("c) Only numbers")
ans = input("Type your answer: ").lower()
if ans == "b":
    score += 1
```

```
print("\n✓ Your Score:", score, "/3")
```

```
# ----- 7. INTERACTIVE GRAPHS -----
```

```
elif choice == "7":
```

```
    while True:
```

```
        print("""
```

```
===== GRAPH OPTIONS =====
```

```
1. Year-wise Financial Loss (range)
```

```
2. Country vs Affected Users
```

```
3. Attack Type Trend
```

```
4. Back
```

```
""")
```

```
    g = input("Type your choice: ").strip()
```

```
    # Graph 1: Year range
```

```
    if g == "1":
```

```
        start = int(input("Start Year: "))
```

```

end = int(input("End Year: "))
sub = df[(df["Year"] >= start) & (df["Year"] <= end)]
loss = sub.groupby("Year")["Financial Loss (in Million $)"].sum()
plt.plot(loss.index, loss.values, marker="o")
plt.title("Financial Loss from " + str(start) + " to " + str(end))
plt.xlabel("Year")
plt.ylabel("Loss (Million $)")
plt.show()

```

Graph 2: Selected countries

```

elif g == "2":
    print("Available countries:")
    print(df["Country"].unique())
    c = input("Type countries separated by comma: ").split(",")
    sub = df[df["Country"].isin([i.strip() for i in c])]
    data = sub.groupby("Country")["Number of Affected Users"].sum()
    data.plot(kind="bar")
    plt.title("Selected Countries vs Affected Users")
    plt.show()

```

Graph 3: Attack Type Trend

```

elif g == "3":
    print("Available attack types:")
    print(df["Attack Type"].unique())
    a = input("Type Attack Type: ")
    sub = df[df["Attack Type"] == a]
    trend = sub.groupby("Year").size()
    plt.plot(trend.index, trend.values, marker='o')
    plt.title("Trend of " + a + " Attacks")
    plt.show()

```

```

elif g == "4":

```

```

    break

```

```

else:

```

```
print("❌ Invalid choice!")
```

```
# ----- 8. COUNTRY BAR GRAPH -----
```

```
elif choice == "8":  
    country_data = df.groupby("Country")["Number of Affected Users"].sum()  
    plt.bar(country_data.index, country_data.values)  
    plt.xticks(rotation=45)  
    plt.title("Country vs Number of Affected Users")  
    plt.show()
```

```
# ----- 9. YEAR vs FINANCIAL LOSS -----
```

```
elif choice == "9":  
    year_loss = df.groupby("Year")["Financial Loss (in Million $)"].mean()  
    plt.plot(year_loss.index, year_loss.values, marker='o')  
    plt.title("Year-wise Average Financial Loss")  
    plt.show()
```

```
# ----- 10. PIE CHART - ATTACK TYPE -----
```

```
elif choice == "10":  
    attack_type = df["Attack Type"].value_counts()  
    plt.pie(attack_type.values, labels=attack_type.index, autopct='%1.1f%%')  
    plt.title("Attack Type Distribution")  
    plt.show()
```

```
# ----- 11. BAR CHART - INDUSTRY -----
```

```
elif choice == "11":  
    industry_data = df["Target Industry"].value_counts()  
    plt.bar(industry_data.index, industry_data.values)  
    plt.xticks(rotation=45)  
    plt.title("Number of Cybersecurity Incidents by Industry")  
    plt.show()
```

```
# ----- 12. TREND ANALYSIS - ATTACK TYPES -----
```

```
elif choice == "12":  
    attack_trend = df.groupby(["Year", "Attack Type"]).size().unstack(fill_value=0)
```

```

attack_trend.plot(kind="line", marker="o")
plt.title("Trend of Attack Types Over Years")
plt.show()

# ----- 13. TREND ANALYSIS - INDUSTRIES -----
elif choice == "13":
    industry_trend = df.groupby(["Year", "Target Industry"]).size().unstack(fill_value=0)
    industry_trend.plot(kind="line", marker="o")
    plt.title("Trend of Target Industry Attacks Over Years")
    plt.show()

# ----- 14. TREND ANALYSIS - FINANCIAL LOSS -----
elif choice == "14":
    financial_trend = df.groupby("Year")["Financial Loss (in Million $)"].sum()
    financial_trend.plot(kind="line", marker='o')
    plt.title("Year-wise Total Financial Loss")
    plt.show()

# ----- 15. PREDICTION -----
elif choice == "15":
    print("\n==== PREDICTION: NEXT YEAR FINANCIAL LOSS ==== \n")

    yearly_loss = df.groupby("Year")["Financial Loss (in Million $)"].sum()
    years = list(yearly_loss.index)
    losses = list(yearly_loss.values)

    if len(losses) < 2:
        print("Not enough data to predict!")
    else:
        increase = losses[-1] - losses[-2]
        next_year = years[-1] + 1
        predicted_loss = losses[-1] + increase

        print("Last year:", years[-1], "Loss:", losses[-1])

```

```

    print("Year before:", years[-2], "Loss:", losses[-2])
    print(f"\nPredicted Financial Loss for {next_year}: {predicted_loss:.2f} Million $")

# ----- 16. OVERALL TREND ANALYSIS -----
elif choice == "16":
    attacks_year = df.groupby("Year").size()
    loss_year = df.groupby("Year")["Financial Loss (in Million $)"].sum()
    plt.plot(attacks_year.index, attacks_year.values, marker='o', label="Total Attacks")
    plt.plot(loss_year.index, loss_year.values, marker='s', label="Financial Loss")
    plt.legend()
    plt.title("Overall Cybersecurity Trends")
    plt.show()

# ----- 17. SUMMARY STATISTICS -----
elif choice == "17":
    print(df.describe(include='all'))

# ----- 18. EXIT -----
elif choice == "18":
    print("✓ Exiting Program...")
    break

else:
    print("✗ Invalid Input!")

```

Output

===== LOGIN REQUIRED =====

Enter Username: admin

Enter Password: 123

✓ Login Successful!

✓ Data Loaded Successfully!

===== MAIN MENU =====

1. Help / Instructions
2. View Sample Data
3. Sorting Options
4. Filter Data
5. Insights / Auto Report
6. Cybersecurity Quiz
7. Interactive Graphs
8. Bar Graph - Country vs Number of Affected Users
9. Line Graph - Year vs Average Financial Loss
10. Pie Chart - Attack Type Distribution
11. Bar Graph - Industry vs Number of Incidents
12. Trend Analysis - Attack Types (Year-wise)
13. Trend Analysis - Industries (Year-wise)
14. Trend Analysis - Financial Loss (Year-wise)
15. Trend Prediction - Next Year Financial Loss
16. Overall Trend Analysis
17. Summary Statistics
18. Exit

Type your choice:

Type your choice: 1

===== HELP / INSTRUCTIONS =====

This project analyzes global cybersecurity data using:

- Menu-driven Python program
- Sorting, Searching, Filtering
- Data Visualization (Graphs)
- Statistical Summary
- Cybersecurity Quiz
- Auto-generated Reports

You can:

- View sample data
- Sort data (high → low or low → high)
- Filter by multiple conditions
- View interactive graphs
- Test your cybersecurity knowledge
- Generate insights/report

=====

Type your choice: 2

===== SAMPLE DATA =====

	Country	Year	Attack Type	Target Industry \
0	China	2019	Phishing	Education
1	China	2019	Ransomware	Retail
2	India	2017	Man-in-the-Middle	IT
3	UK	2024	Ransomware	Telecommunications
4	Germany	2018	Man-in-the-Middle	IT
5	Germany	2017	Man-in-the-Middle	Retail
6	Germany	2016	DDoS	Telecommunications
7	France	2018	SQL Injection	Government
8	India	2016	Man-in-the-Middle	Banking
9	UK	2023	DDoS	Healthcare

	Financial Loss (in Million \$)	Number of Affected Users	Attack Source \
0	80.53	773169	Hacker Group
1	62.19	295961	Hacker Group
2	38.65	605895	Hacker Group
3	41.44	659320	Nation-state
4	74.41	810682	Insider
5	98.24	285201	Unknown
6	33.26	431262	Insider
7	59.23	909991	Unknown
8	16.88	698249	Unknown
9	69.14	685927	Hacker Group

	Security Vulnerability Type	Defense Mechanism Used \
0	Unpatched Software	VPN
1	Unpatched Software	Firewall
2	Weak Passwords	VPN
3	Social Engineering	AI-based Detection
4	Social Engineering	VPN
5	Social Engineering	Antivirus
6	Unpatched Software	VPN
7	Social Engineering	Antivirus
8	Social Engineering	VPN
9	Unpatched Software	Firewall

	Incident Resolution Time (in Hours)	Country_clean	Attack_clean \
0	63	china	phishing
1	71	china	ransomware
2	20	india	man-in-the-middle
3	7	uk	ransomware
4	68	germany	man-in-the-middle
5	25	germany	man-in-the-middle
6	34	germany	ddos
7	66	france	sql injection
8	47	india	man-in-the-middle
9	58	uk	ddos

	Industry_clean	Year_clean
0	education	2019
1	retail	2019
2	it	2017
3	telecommunications	2024
4	it	2018
5	retail	2017
6	telecommunications	2016
7	government	2018
8	banking	2016
9	healthcare	2023

Type your choice: 3

===== SORTING MENU =====

1. Sort by Financial Loss (High → Low)
2. Sort by Affected Users (High → Low)
3. Sort by Year (Old → New)

Type your choice:

Type your choice: 1

	Country	Year	Attack Type	Target Industry \
2030	China	2024	DDoS	Banking
1806	Australia	2017	SQL Injection	IT
2133	Germany	2016	Phishing	Telecommunications
419	Germany	2019	Phishing	Healthcare
1363	Brazil	2020	Ransomware	Banking
2004	USA	2016	Ransomware	Telecommunications
2092	Germany	2015	Ransomware	Telecommunications
2228	Japan	2020	DDoS	Education
2428	USA	2021	DDoS	Telecommunications
1678	USA	2020	DDoS	Banking

	Financial Loss (in Million \$)	Number of Affected Users	Attack Source \
2030	99.99	755185	Unknown
1806	99.99	672966	Hacker Group
2133	99.98	761576	Insider
419	99.97	950451	Hacker Group
1363	99.90	429566	Nation-state
2004	99.88	999508	Unknown
2092	99.83	419344	Hacker Group
2228	99.83	10859	Nation-state
2428	99.81	460842	Unknown
1678	99.81	948174	Unknown

	Security Vulnerability Type	Defense Mechanism Used	\
2030	Weak Passwords	Encryption	
1806	Zero-day	Firewall	
2133	Unpatched Software	Antivirus	
419	Weak Passwords	AI-based Detection	
1363	Zero-day	AI-based Detection	
2004	Unpatched Software	VPN	
2092	Zero-day	Firewall	
2228	Social Engineering	AI-based Detection	
2428	Weak Passwords	AI-based Detection	
1678	Social Engineering	AI-based Detection	

	Incident Resolution Time (in Hours)	Country_clean	Attack_clean	\
2030	20	china	ddos	
1806	13	australia	sql injection	
2133	9	germany	phishing	
419	29	germany	phishing	
1363	54	brazil	ransomware	
2004	23	usa	ransomware	
2092	59	germany	ransomware	
2228	66	japan	ddos	
2428	59	usa	ddos	
1678	4	usa	ddos	

	Industry_clean	Year_clean
2030	banking	2024
1806	it	2017
2133	telecommunications	2016
419	healthcare	2019
1363	banking	2020
2004	telecommunications	2016
2092	telecommunications	2015
2228	education	2020
2428	telecommunications	2021
1678	banking	2020

===== SORTING MENU =====

1. Sort by Financial Loss (High → Low)
2. Sort by Affected Users (High → Low)
3. Sort by Year (Old → New)

Type your choice: 2

	Country	Year	Attack Type	Target Industry	\
2045	USA	2023	Ransomware	Government	
323	France	2022	SQL Injection	Healthcare	
2240	India	2020	Malware	IT	
2004	USA	2016	Ransomware	Telecommunications	
2440	USA	2020	DDoS	Banking	
1515	India	2024	Ransomware	Government	
2852	China	2021	Ransomware	Retail	
2775	UK	2023	Man-in-the-Middle	Banking	
928	France	2019	Phishing	Retail	
1561	USA	2016	Phishing	Retail	

	Financial Loss (in Million \$)	Number of Affected Users	Attack Source	\
2045	93.34	999635	Nation-state	
323	27.06	999545	Hacker Group	
2240	84.43	999542	Hacker Group	
2004	99.88	999508	Unknown	
2440	60.20	998937	Nation-state	
1515	79.30	998833	Hacker Group	
2852	40.01	998728	Unknown	
2775	24.28	998547	Unknown	
928	25.75	998507	Insider	
1561	89.85	998085	Insider	

	Security Vulnerability Type	Defense Mechanism Used	\
2045	Weak Passwords	Firewall	
323	Social Engineering	Encryption	
2240	Social Engineering	Encryption	
2004	Unpatched Software	VPN	
2440	Zero-day	Antivirus	
1515	Social Engineering	Firewall	
2852	Weak Passwords	AI-based Detection	
2775	Weak Passwords	Antivirus	
928	Weak Passwords	AI-based Detection	
1561	Zero-day	VPN	

	Incident Resolution Time (in Hours)	Country_clean	Attack_clean	\
2045	24	usa	ransomware	
323	47	france	sql injection	
2240	15	india	malware	
2004	23	usa	ransomware	
2440	19	usa	ddos	
1515	21	india	ransomware	
2852	68	china	ransomware	
2775	46	uk	man-in-the-middle	
928	15	france	phishing	
1561	7	usa	phishing	

	Industry_clean	Year_clean
2045	government	2023
323	healthcare	2022
2240	it	2020
2004	telecommunications	2016
2440	banking	2020
1515	government	2024
2852	retail	2021
2775	banking	2023
928	retail	2019
1561	retail	2016

Type your choice: 3

===== SORTING MENU =====

1. Sort by Financial Loss (High → Low)
2. Sort by Affected Users (High → Low)
3. Sort by Year (Old → New)

Type your choice: 5

✗ Invalid Choice

Type your choice: 4

===== FILTER MENU =====

1. Industry & Attack Type Filter
2. Country-wise Yearly Summary
3. Back to Main Menu

Type your choice:

Type your choice: 4

===== FILTER MENU =====

1. Industry & Attack Type Filter
2. Country-wise Yearly Summary
3. Back to Main Menu

Type your choice: 1

Available Attack Types:

['DDoS', 'Malware', 'Man-in-the-Middle', 'Phishing', 'Ransomware', 'SQL Injection']

Available Industries:

['Banking', 'Education', 'Government', 'Healthcare', 'IT', 'Retail', 'Telecommunications']

Type Attack Type (or press Enter to skip): ddos

Type Industry (or press Enter to skip): it

✓ FILTERED DATA:

	Country	Year	Attack Type	Target Industry	\
14	France	2024	DDoS	IT	
17	India	2015	DDoS	IT	
19	India	2016	DDoS	IT	
43	USA	2017	DDoS	IT	
70	Brazil	2015	DDoS	IT	
...	...	...	...	...	
2867	USA	2023	DDoS	IT	
2868	Australia	2024	DDoS	IT	
2893	Australia	2021	DDoS	IT	
2917	Germany	2019	DDoS	IT	
2927	USA	2022	DDoS	IT	

	Financial Loss (in Million \$)	Number of Affected Users
14	48.01	922258
17	93.14	805278
19	36.45	261808
43	74.12	403108
70	43.51	940656
...	...	...
2867	55.14	529551
2868	82.49	102295
2893	47.83	126709
2917	71.08	445129
2927	74.59	281698

[91 rows x 6 columns]

===== FILTER MENU =====

1. Industry & Attack Type Filter
2. Country-wise Yearly Summary
3. Back to Main Menu

Type your choice: 2

Available Countries:

['Australia', 'Brazil', 'China', 'France', 'Germany', 'India', 'Japan', 'Russia', 'UK', 'USA']

Available Years:

[2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024]

Type Country (or press Enter to skip): india

Type Year (or press Enter to skip): 2017

✓ FILTERED DATA:

	Country	Year	Attack Type	Target Industry \
2	India	2017	Man-in-the-Middle	IT
22	India	2017	DDoS	Government
192	India	2017	Man-in-the-Middle	Government
269	India	2017	Phishing	Retail
327	India	2017	Malware	Telecommunications
453	India	2017	DDoS	Government
489	India	2017	DDoS	IT
597	India	2017	Ransomware	Retail
633	India	2017	Ransomware	Retail
640	India	2017	Ransomware	Banking
689	India	2017	SQL Injection	IT
716	India	2017	Phishing	Retail
809	India	2017	DDoS	Education
910	India	2017	Ransomware	Education
929	India	2017	DDoS	Retail

2644	India	2017	Malware	Healthcare
	Financial Loss (in Million \$)		Number of Affected Users	
2			38.65	605895
22			76.71	246205
192			69.69	713419
269			36.78	620056
327			7.44	552358
453			75.52	352250
489			70.09	417831
597			80.93	837506
633			69.91	340928
640			9.81	489620
689			78.94	858565
716			97.87	113745
809			83.49	79933
910			98.11	677198
929			82.17	593329
1211			20.21	421023
2550			53.95	920801
2644			66.29	430200
===== FILTER MENU =====				
1. Industry & Attack Type Filter				
2. Country-wise Yearly Summary				
3. Back to Main Menu				
Type your choice: 3				
Returning to Main Menu...				
===== MAIN MENU =====				
1. Help / Instructions				
2. View Sample Data				
3. Sorting Options				
4. Filter Data				
5. Insights / Auto Report				
6. Cybersecurity Quiz				
7. Interactive Graphs				
8. Bar Graph - Country vs. Number of Affected Users				

-
13. Trend Analysis - Industries (Year-wise)
 14. Trend Analysis - Financial Loss (Year-wise)
 15. Trend Prediction - Next Year Financial Loss
 16. Overall Trend Analysis
 17. Summary Statistics
 18. Exit

Type your choice: 5

```
===== CYBERSECURITY INSIGHTS REPORT =====
Most Attacked Country: Brazil
Most Common Attack Type: DDoS
Year with Most Attacks: 2017
Year with Highest Financial Loss: 2017
=====
```

```
===== MAIN MENU =====
1. Help / Instructions
2. View Sample Data
```

Type your choice: 6

```
===== CYBERSECURITY QUIZ =====
```

1) What is Phishing?
a) Fake emails to steal data
b) Hardware failure
c) Strong password
Type your answer: a

2) What does Ransomware do?
a) Displays ads
b) Locks your data for ransom
c) Cleans virus
Type your answer: a

3) Strong password includes?
a) Only letters

2) What does Ransomware do?
a) Displays ads
b) Locks your data for ransom
c) Cleans virus
Type your answer: a

3) Strong password includes?
a) Only letters
b) Letters, numbers, symbols
c) Only numbers
Type your answer: b

✓ Your Score: 2 /3

===== MAIN MENU =====

1. Help / Instructions
2. View Sample Data
3. Sorting Options

Type your choice: 7

===== GRAPH OPTIONS =====

1. Year-wise Financial Loss (range)
2. Country vs Affected Users
3. Attack Type Trend
4. Back

Type your choice: 1
Start Year: 2015
End Year: 2022

===== GRAPH OPTIONS =====

1. Year-wise Financial Loss (range)
2. Country vs Affected Users
3. Attack Type Trend
4. Back

Type your choice: 2

Available countries:

['China' 'India' 'UK' 'Germany' 'France' 'Australia' 'Russia' 'Brazil'
'Japan' 'USA']

Type countries separated by comma:

===== GRAPH OPTIONS =====

1. Year-wise Financial Loss (range)
2. Country vs Affected Users
3. Attack Type Trend
4. Back

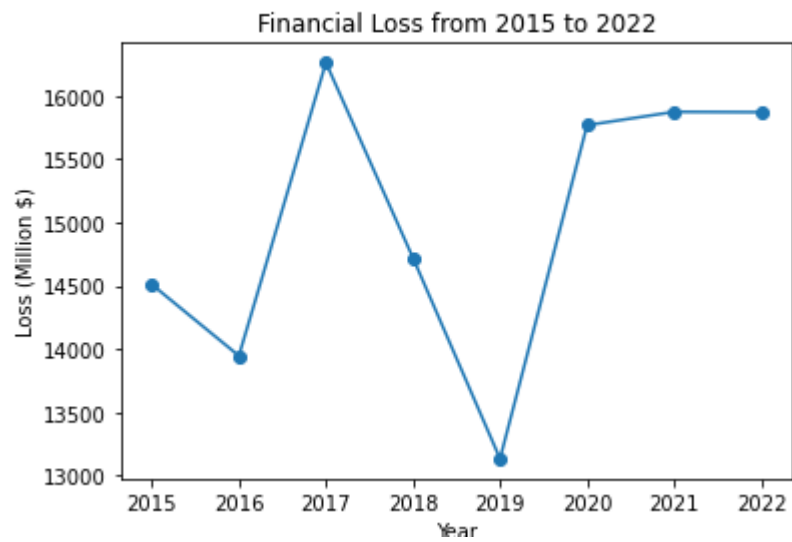
Type your choice: 3

Available attack types:

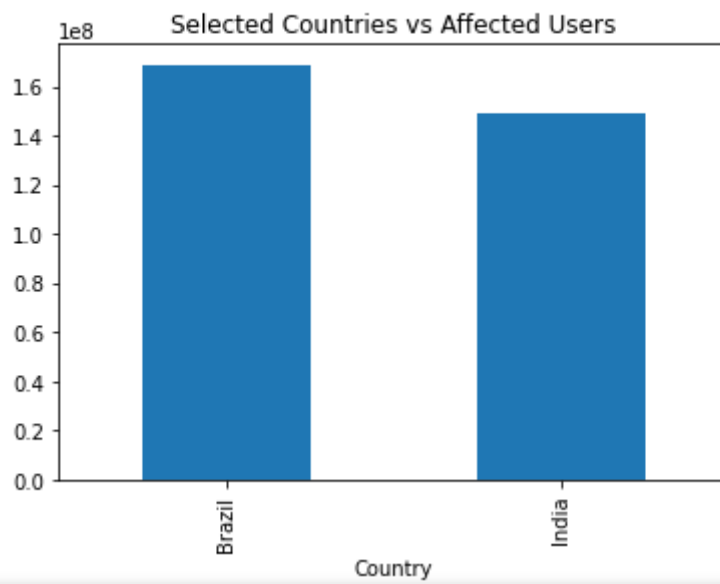
['Phishing' 'Ransomware' 'Man-in-the-Middle' 'DDoS' 'SQL Injection'
'Malware']

Type Attack Type:

End Year: 2022



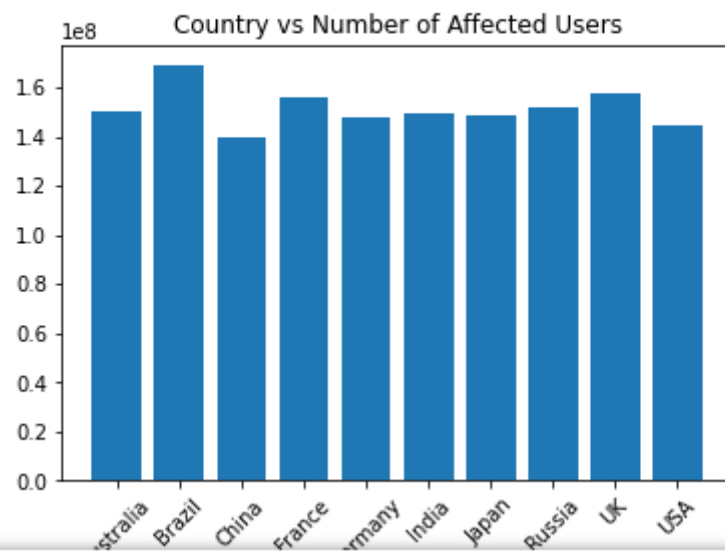
Type countries separated by comma: brazil, india



Type Attack Type: phishing



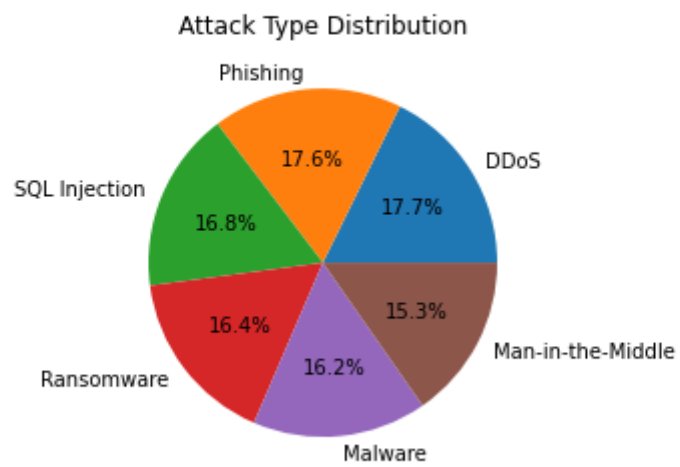
Type your choice: 8



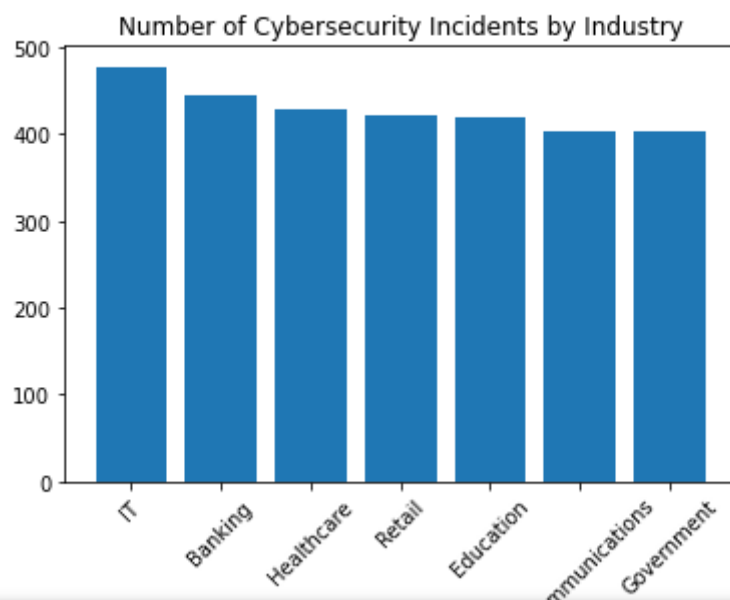
Type your choice: 9



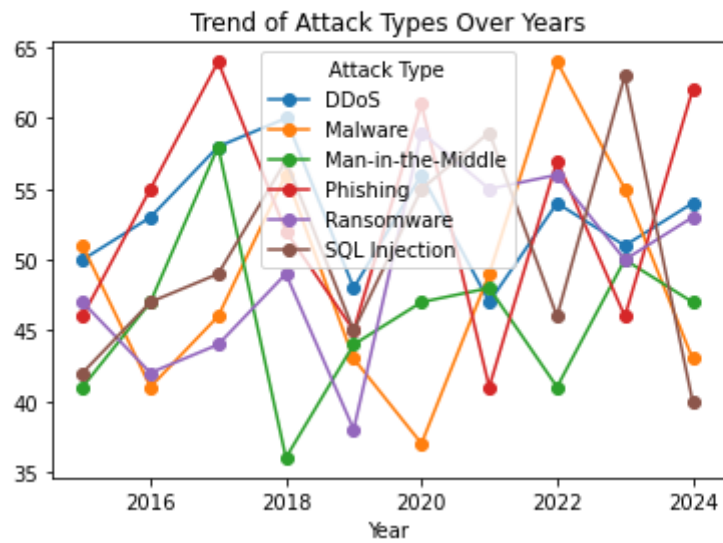
Type your choice: 10



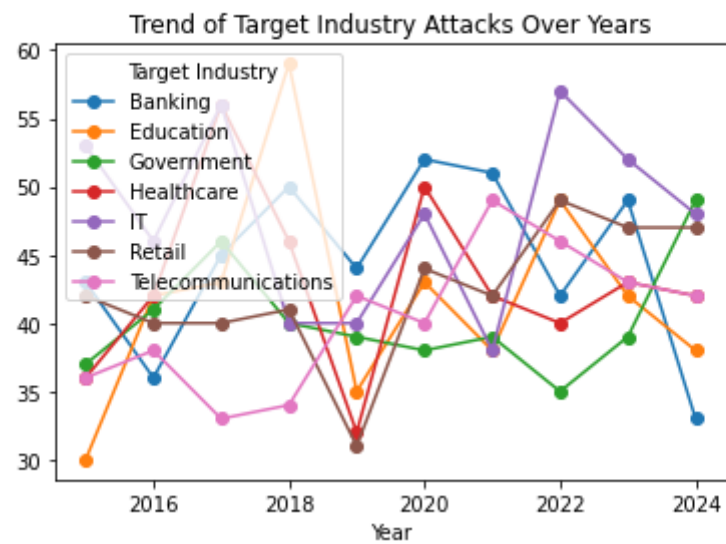
Type your choice: 11



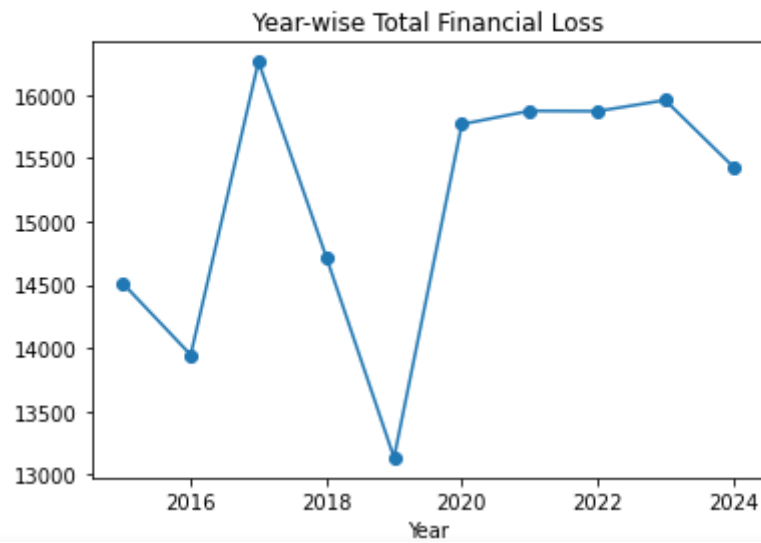
Type your choice: 12



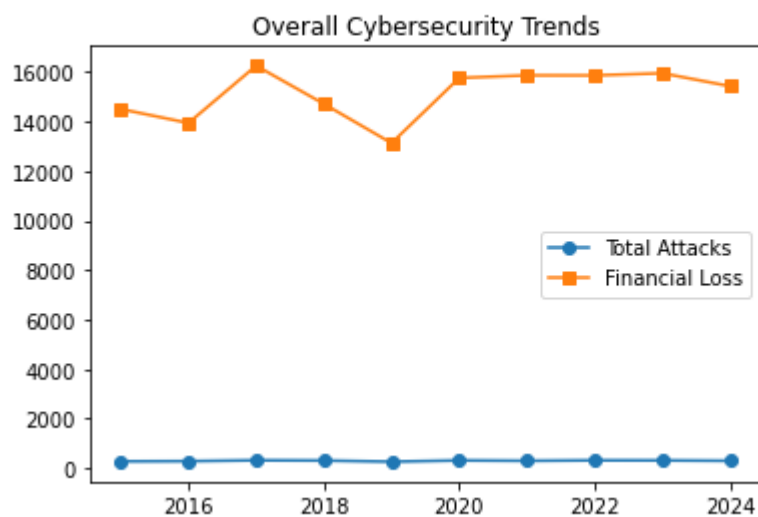
Type your choice: 13



Type your choice: 14



Type your choice: 16



Type your choice: 15

==== PREDICTION: NEXT YEAR FINANCIAL LOSS ====

Last year: 2024 Loss: 15434.29

Year before: 2023 Loss: 15958.08

Predicted Financial Loss for 2025: 14910.50 Million \$

Type your choice: 17

	Country	Year	Attack Type	Target Industry	\
count	3000	3000.000000	3000	3000	
unique	10	NaN	6	7	
top	UK	NaN	DDoS	IT	
freq	321	NaN	531	478	
mean	NaN	2019.570333	NaN	NaN	
std	NaN	2.857932	NaN	NaN	
min	NaN	2015.000000	NaN	NaN	
25%	NaN	2017.000000	NaN	NaN	
50%	NaN	2020.000000	NaN	NaN	
75%	NaN	2022.000000	NaN	NaN	
max	NaN	2024.000000	NaN	NaN	

	Financial Loss (in Million \$)	Number of Affected Users	Attack Source	\
count	3000.000000	3000.000000	3000	
unique	NaN	NaN	4	
top	NaN	NaN	Nation-state	
freq	NaN	NaN	794	
mean	50.492970	504684.136333	NaN	
std	28.791415	289944.084972	NaN	
min	0.500000	424.000000	NaN	
25%	25.757500	255805.250000	NaN	
50%	50.795000	504513.000000	NaN	
75%	75.630000	758088.500000	NaN	
max	99.990000	999635.000000	NaN	

	Security Vulnerability Type	Defense Mechanism Used	\
count	3000	3000	
unique	4	5	
top	Zero-day	Antivirus	
freq	785	628	
mean	NaN	NaN	
std	NaN	NaN	
min	NaN	NaN	
25%	NaN	NaN	
50%	NaN	NaN	
75%	NaN	NaN	
max	NaN	NaN	

	Incident Resolution Time (in Hours)	Country_clean	Attack_clean	\
count	3000.000000	3000	3000	
unique	NaN	10	6	
top	NaN	uk	ddos	
freq	NaN	321	531	
mean	36.476000	NaN	NaN	
std	20.570768	NaN	NaN	
min	1.000000	NaN	NaN	
25%	19.000000	NaN	NaN	
50%	37.000000	NaN	NaN	
75%	55.000000	NaN	NaN	
max	72.000000	NaN	NaN	

	Industry_clean	Year_clean
count	3000	3000.000000
unique	7	NaN
top	it	NaN
freq	478	NaN
mean	NaN	2019.570333
std	NaN	2.857932
min	NaN	2015.000000
25%	NaN	2017.000000
50%	NaN	2020.000000
75%	NaN	2022.000000
max	NaN	2024.000000

Type your choice: 18

✓ Exiting Program...

Security Control

1. Login authentication implemented using username and password.
2. Data cleaning ensures safe filtering and prevents invalid queries.
3. Error handling prevents crashes due to incorrect input.
4. Restricted access – Program exits if login fails.
5. The modular menu system ensures controlled navigation.

Future Scope of the Project

In the future, this project can be enhanced by using real-time cybersecurity data, integrating machine learning models to predict potential threats, and creating interactive dashboards for better visualization.

Additional features such as automated alert systems, live attack tracking, and deeper analysis of emerging cyberattack trends can make the project more advanced and useful.

The system can also be expanded to cover larger datasets and support cloud-based security monitoring for improved performance and accessibility.

Overall:

1. Implement machine learning for advanced prediction.
2. Add database integration instead of CSV.
3. Create a GUI-based dashboard.
4. Implement real-time cybersecurity monitoring using APIs.

Conclusion

This project provided a meaningful understanding of cybersecurity threats by analyzing data using Python.

Through the use of CSV files, data processing techniques, and visualizations such as bar graphs, line graphs, and pie charts, the project successfully highlighted patterns, trends, and the impact of various cyberattacks.

It also helped in developing important skills in programming, data handling, and interpretation of charts.

The project achieved its aim of spreading awareness about cybersecurity and demonstrated how data analysis can be used to better understand and address digital threats in today's world.

Overall, this project successfully demonstrates cyber threat analysis using Python.

It includes menu-driven interaction, advanced filtering, graphical insights, trend analysis, and prediction modeling.

The structured and secure design makes it ideal for educational and practical use.

Bibliography

- CBSE Informatics Practices Textbook
- Python Documentation